

Hamza Zia

Machine Learning Engineer | Data Scientist

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Education

Massachusetts Institute of Technology (MIT)
MicroMasters Statistics and Data Science

Jan 2025 - Present

Virtual University of Pakistan
Bachelor of Science Computer Science

Apr 2019 - Apr 2023

Work Experience

Data Science & Machine Learning Trainee
Dice Analytics

Feb 2025 – Apr 2025

Machine Learning Fundamentals, Data Ingestion/ETL, Python Programming, Data Wrangling Expertise, Exploratory Data Analysis (EDA), Data Communication, Data Exploration, Problem identification

Convergent Graduate Academic Program (CGAP)
Convergent Business Technologies

Aug 2023 - Sep 2023

- CS50: Introduction to Computer Science by HarvardX.
- Gained foundational knowledge in programming, algorithms, data structures, and computer science principles.
- Worked on problem-solving exercises and coding assignments to strengthen computational thinking.

Project Intern

Global Consulting Services

May 2023 - Jun 2023

- Learned and worked with BIRT (Business Intelligence and Reporting Tools) for report generation.
- Gained hands-on experience in SQL for querying and managing databases.
- Explored IBM Maximo, understanding its role in enterprise asset management.

Projects

Deepfake Detection Using Deep Learning (FYP)

Model can predict a fake video generated using (A.I) deep learning with 85% Recall. Model has the ability to detect voice and face swap

Customer Segmentation Based On Annual Spending

Model can classify with precision of 95% that a customer belongs to a specific category, it is further used for recommending products, deals, promotions.

Employee Turnover Monitoring Model

Model can successfully predict with 97% accuracy based on several features such as promoted in last 5 years, pay, weekly hours etc. whether the employee will leave the company or not.

Customised Train Ticket Price Predictor

Price of the ticket can be predicted with 94% precision, pricing is changed regularly based on demand and time. Data is scraped regularly and it can be predicted what the fare will be given destination, origin, class of train etc.

Core Skills

- Machine Learning
- Tensorflow
- Scikit-learn
- Python
- Power BI
- Numpy
- Pandas
- Python RegEx
- Git
- SQL
- HTML
- CSS
- BIRT (Business Intelligence and Reporting Tools)
- Snap MI

Certifications & Trainings

Kaggle

Introduction to Deep Learning

Stanford University

Generative AI for Everyone

Machine Learning Specialization

Supervised Machine Learning: Regression and Classification

Advanced Learning Algorithms

Unsupervised Learning, Recommenders, Reinforcement Learning

Harvard University

CS50's Introduction to Programming with Python

CS50's Introduction to Computer Science

Dice Analytics

Data Science and Machine Learning

****Links to certifications & notebooks are embedded in the titles****